

Full Length Article

Optical image analysis for graphene layer detection: Enhanced green channel methodology

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ABSTRACT

Graphene, a material of increasing research interest, requires accurate layer identification due to its sensitivity to layer count. Existing methods for graphene layer number identification are either time-consuming or of low accuracy, with high-accuracy methods often requiring expensive processes. This paper aims to address this challenge by proposing a cost-effective and efficient approach. Specifically, the current work highlights only the green channel—one of the three primary color channels (red, green, blue) that make up an optical image—from images of exfoliated graphene flakes for layer count identification. A linear regression is performed between pixel position and substrate green channel value, and this effect is subtracted from the entire optical image to mitigate background effects. By storing the range of green channel values for each type of flake (monolayer, bilayer, or tri-layer) based on a few images, we establish thresholds for identifying different types of layers in a particular setup. Additionally, our methodology allows for flexible threshold tuning using a single reference image, enabling adjustment to changes in detection setup such as illumination level, magnification, or microscope used. Demonstrating high accuracy and flexibility, this methodology presents a suitable technique for graphene layer number identification without the need for large datasets or expensive instruments.

1. Introduction

Graphene is an interesting material by virtue of its exceptional mechanical, thermal, optical, and electrical properties, especially for exploring exotic physics. These properties have widened the usage of graphene in various applications, including energy storage, conducting electrodes, sensors, light-emitting diodes (LEDs), photovoltaic cells, and numerous other applications [1,2]. While there is on-going effort on large-area synthesis—chemical vapor deposition (CVD) [3], metal–organic chemical vapor deposition (MOCVD) [4], molecular beam epitaxy (MBE) [5], exfoliation is the only method for producing high-quality graphene [6]. The characteristic properties of exfoliated graphene are heavily influenced by the 2D structure, classified based on the layer count: monolayer, bilayer, tri-layer, and few-layer graphene. Although monolayers exhibit diverse applications, bilayers, or few-layers, are favored for materials with enhanced thermal conduction

[7], strength [8], and chemical functionality [9]. Furthermore, addressing each layer separately introduces new aspects of electrical properties, including controlling the band gap through doping or gating [10–14]. Consequently, accurately determining the layer count is vital to thoroughly examining the characteristic features of multilayer graphene.

There are two distinct efforts—heuristic and computational methods to determine the layer count. The heuristic approach often involves experimentation with the parameters and conditions of various practical techniques such as Raman spectroscopy, scanning electron microscopy (SEM), and atomic force microscopy (AFM) [15–17]. With Raman Spectroscopy, by simply measuring the shift in spectral peaks, one can measure the graphene layer thicknesses from a monolayer to 10 layers quite rapidly. While this is true, using optical images together with a versatile image recognition and analysis package (presented in this work) offers a powerful, quick, and accurate alternative. Generic

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0927-0256/© 2024 Elsevier B.V. All rights reserved, including those for text and data mining, AI training, and similar technologies.

computational technologies for 2D materials detection have progressed significantly offering fast and efficient detection of graphene layers. However, these methods also have certain limitations.

Recent research by F. Obelenis and A. Champi utilized grayscale image processing technique to count the number of graphene monolayers leveraging optical microscopic images [18]. However, this method is dependent on the calibration of equipment utilizing various physical methods (i.e., AFM and Raman). Deep-learning-based image segmentation algorithms have also been explored, which enable precise recognition of 2D materials instantaneously through optical microscope pictures, leveraging high-dimensional hierarchical visual properties [19]. Furthermore, various 2D materials analogous to graphene, such as transition metal dichalcogenide (TMDC) and hexagonal boron nitride (hBN), can also be successfully identified using a neural network-based approach like Mask-CNN [20]. Nevertheless, these techniques require a great amount of image data handling and preparation and are computationally prohibitive. Moreover, obtaining extensive sets of high-quality optical image data typically proves to be a challenging task. As the graphene research field progresses, it is imperative to stay updated with these identification methodologies, enabling more refined and detailed characterizations that are pivotal for the development of graphene-based materials and devices across various industries.

Digital images are composed of pixels, and each pixel is made up of a combination of three primary colors: red, green, and blue. These color components are referred to as color channels and are known as red channel, green channel, and blue channel (RGB). This group recently published a successful breakthrough in optical image processing of exfoliated graphene layers on SiO₂ substrates under controlled illumination utilizing a novel green channel correlation thresholding technique [21]. Manual cross-checking and pixel accuracy ratio results from the proposed green channel correlation method demonstrated excellent accuracy given that the optical images were taken under stable lighting conditions. One of the pillars of success of the work rested on the computationally inexpensive nature of the segmentation technique to identify the various layers in a given optical image of a graphene flake. This was much faster and more efficient compared to existing computational methods.

Building upon the highlighted prior work, the current study enhances the green channel correlation thresholding method, more streamlining the process by providing threshold tuning mechanism, eliminating the need for controlled lighting conditions. This technique is capable of fine-tuning the threshold values based on a reference image via a graphical user interface (GUI) program, ensuring segmentation adaptability to varying illumination conditions. Furthermore, it is capable of identifying the nuanced variations in contrast between the background and the graphene flakes. It underscores the algorithm's capacity for detecting even the most intricate details within the images. In comparison to conventional methods, this methodology uses less data and can produce excellent results. Because of its versatility, it can be used in a range of settings with different lighting conditions, outside interference, and magnification levels. Its computing efficiency further qualifies it for real-world applications, negating the need for large datasets or extensive preprocessing.

2. Methodology

At the onset, it is important to note the basic optics of the setup to generate images. A conventional optical microscope was used to generate the RGB channel. The graphene flakes (of which the optical images were taken) rested on a conventional silicon wafer with a 285-nanometer-thick oxide layer. Regarding the approximate size of these flakes, the flake sizes vary. Scale bars have been added to Figs. 5–8 to showcase accurate image dimensions.

The first step in the image analyses is the acquisition of an initial set of five reference images for each category of graphene flakes, namely monolayer, bilayer, and tri-layer. The images undergo processing to

enhance their quality, following which thresholds are derived based on the green channel value. These thresholds are then stored and utilized for further identification.

2.1. Image processing

Initially, the raw image undergoes processing steps aimed at enhancing its quality. To reduce noise while preserving flake edge sharpness, a bilateral filter (Supplementary information) is applied to the image. Subsequently, an automated approach based on histogram analysis is employed to determine the green pixel value corresponding to the background, or substrate, of the image. Pixels with green channel values exceeding 160, typically associated with non-background elements, are filtered out. The remaining pixels' green channel values are then used to construct a histogram, wherein peak values are identified and sorted. The peak with the highest green channel value is indicative of the background's green channel value (denoted as G_{back}). This determination is logical as backgrounds tend to exhibit lighter tones, thereby resulting in higher green channel values. Following this, pixels satisfying the condition $G_{back} - 1 \leq G_{px} \leq G_{back} + 1$ are flagged as background pixels, and their coordinates are recorded. These coordinates are used to derive a linear equation of the form:

$$G_{px} = aX + bY + c \quad (1)$$

using the method of least squares. Here, X and Y denote the pixel's X and Y coordinates, respectively, while G represents its green channel value. The intercept term is represented by ' c '. It provides a baseline value for the dependent variable G_{px} when all the features are zero. Equation (1) establishes a relationship between a pixel's position and its green channel value. Utilizing this equation, the green channel value for each pixel across the image is recalculated and subtracted from its actual green channel value. This corrective measure effectively diminishes the background's influence on graphene flakes.

2.2. Thresholding

After processing all the images, pixel values were systematically extracted from various regions within the flakes present in the processed images. For every individual flake, the median values for each color channel (red, green, and blue) were computed. These derived median pixel values were subsequently employed in the creation of graphical representations. Along the Y -axis, the median pixel value was plotted, while the X -axis represented the number of graphene layers, depicted in Fig. 1(b-d). The resulting graphs revealed a noticeable relationship among the median red, green, and blue pixel values.

A closer examination of these graphs shows that although green channel and red channel values vary with layer number, this variation is not significant for blue channel values. Overall, the following relation becomes evident— $G_{mono} > G_{bi} > G_{tri}$ and $R_{mono} > R_{bi} > R_{tri}$ where G and R represents the average green channel and red channel value of different layers. So, blue channel values are not considered for thresholding. Furthermore, analyzing the standard deviation, it becomes evident that the deviation of green channel values is less than the deviation of red channel values (Fig. 2(a-c)).

Given these results, green channel is therefore chosen for threshold development. With the green channel value selected for thresholding, Fig. 1(a) was utilized for the development of thresholds. The upper threshold was determined by selecting the highest value of the green channel from the graph, while the lower threshold was set based on the lowest value. This process was independently executed for all three layers, resulting in a complete set of thresholds ready for the segmentation of new images. The thresholds for monolayer, bilayer and tri-layer are denoted as T_{mono} , T_{bi} and T_{tri} consecutively. Fig. 3 visually demonstrates the thresholds. These thresholds are then stored for further layer identification.

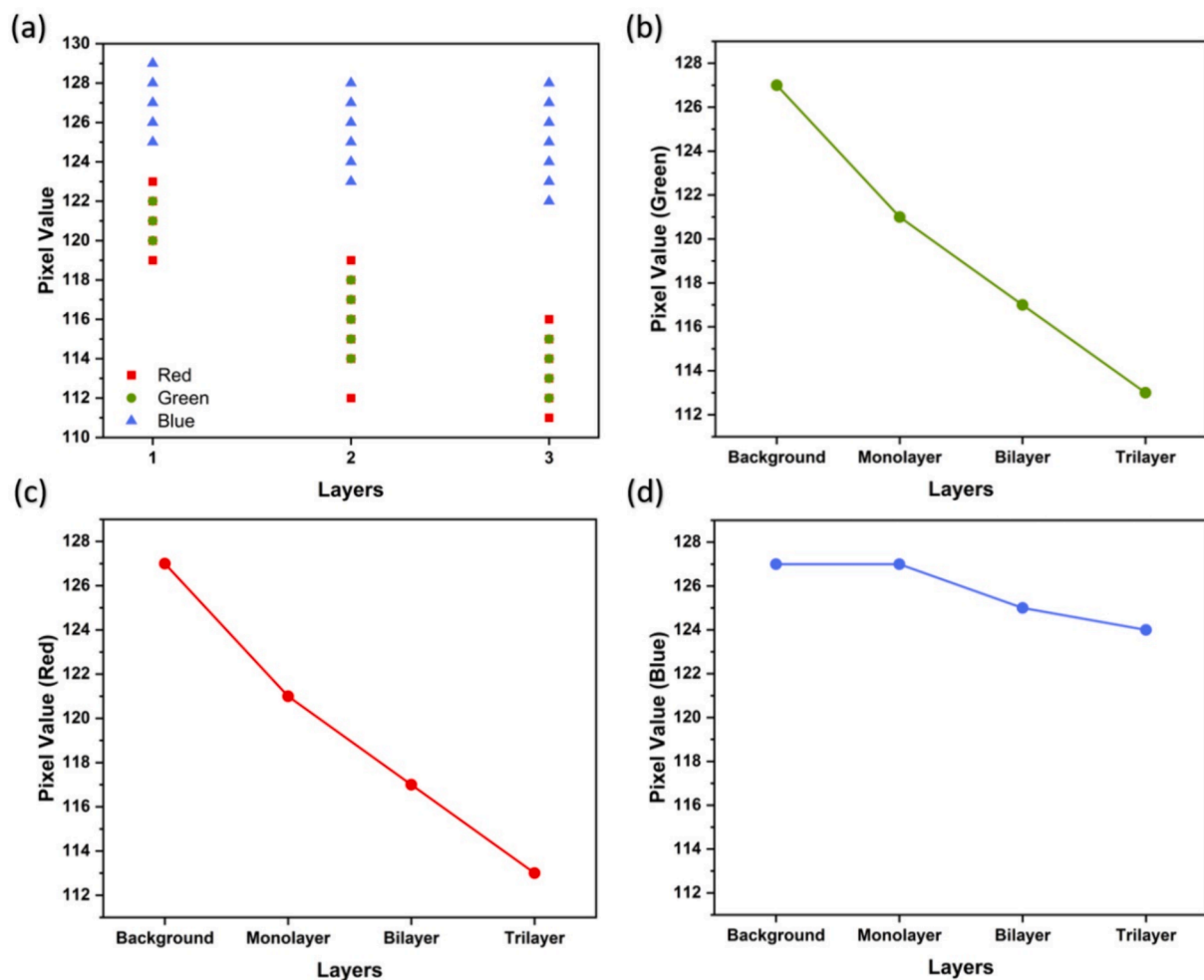


Fig. 1. (a) The range of red, green, and blue channel for all layers in a total of 15 images. (b–d) Median values of the green, red, and blue channels for each layer.

2.3. Identification

When a new raw optical image is inputted, it undergoes the same image processing steps as mentioned previously. The stored threshold values are retrieved and utilized to identify layers in the new image. Pixels with green channel values falling within specific ranges are classified as monolayer, bilayer, or tri-layer graphene. The identified layers are then displayed in the image in different colors (Fig. 4).

3. Results and discussion

Fig. 5 illustrates the progression of the image processing stages, displaying the initial image, the filtered image, and the segmented images corresponding to monolayer, bilayer, and tri-layer graphene.

The segmented images exhibit a clear segmentation of the various graphene layers, demonstrating the effectiveness of this methodology. However, it's worth noting that the presence of uncolored regions within the segmented images may arise from either false negative conditions or true positive classifications. These areas warrant further investigation to better understand the specific conditions under which they occur.

3.1. Evaluation method

Numerous recent studies have used the pixel accuracy ratio for evaluation purposes. However, this metric can run into issues with class imbalance. Class imbalance occurs when some classes have way more

instances than others. For example- in images containing graphene flakes, the majority of the image might consist of background or other non-flake elements, while the actual flakes themselves are relatively rare. Due to this extreme class imbalance, even if the algorithm correctly identifies most of the background pixels but struggles to detect the rare flakes, the accuracy metric can still show very high scores. Therefore, to evaluate the accuracy of segmented images the intersection over union (*IoU*) method is chosen, which is more appropriate for datasets with imbalanced classes. The general formula for *IoU* can be seen in Eq. (2).

$$IoU = \frac{TP}{TP + FN + FP} \quad (2)$$

Here, *FP* (False Positive) signifies instances of incorrect identification of the positive group, for example, pixels erroneously assigned as monolayer pixels outside the monolayer flake. *FN* (False Negative) represents cases where the test results incorrectly indicate the absence of a condition that is, in fact, present, such as failing to identify pixels as monolayer within a monolayer flake. *TP* (True Positive) denotes the correct identification of the positive group, like identifying pixels as monolayer within a monolayer flake.

The accuracy calculated using the *IoU* approach for monolayer and bilayer flakes has been succinctly summarized in Table 1. It is to note that, bilayer and tri-layer graphene layers exhibit the edge effect, a phenomenon elaborated on in the subsequent section. When assessing accuracy, even though the edge is consistently recognized as a monolayer, it is still considered a true positive.

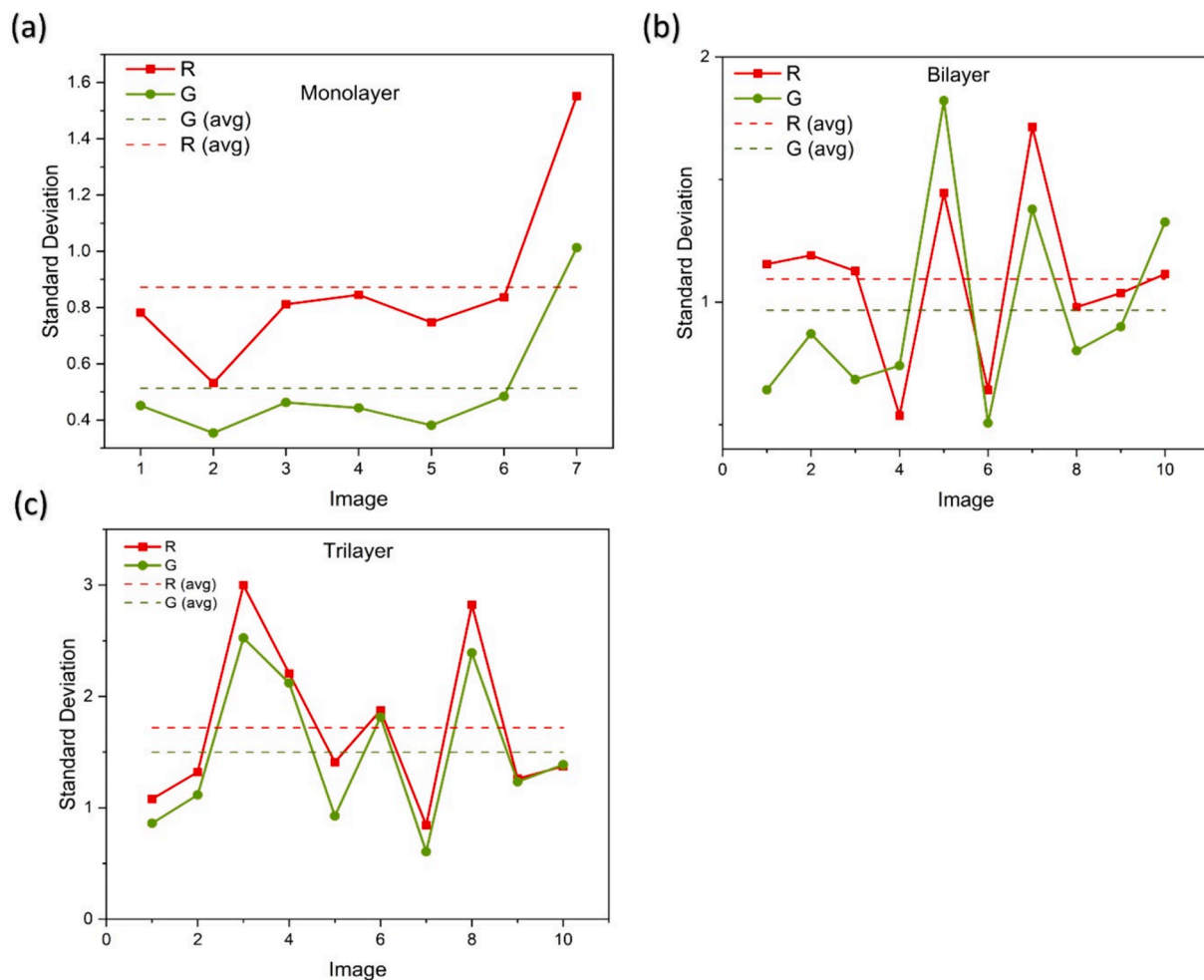


Fig. 2. Comparison of the standard deviation of each channel for (a) monolayer, (b) bilayer and (c) tri-layer.

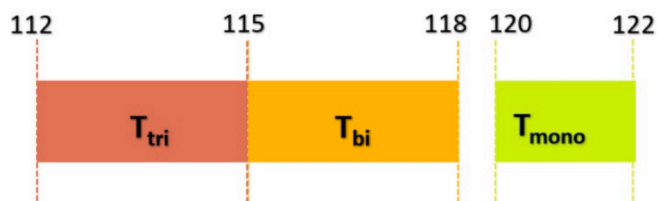


Fig. 3. Visual representation of thresholds for each layer.

3.2. Edge effect

During the segmentation process, an interesting phenomenon was observed, referred to as the 'edge effect.' This phenomenon involves the identification of monolayer graphene edges as part of the monolayer material by the segmentation algorithm. However, in the case of bilayer and tri-layer graphene, the same algorithm tends to classify the edges as monolayer graphene. This peculiar behavior can be attributed to subtle variations in the contrast between the background and the graphene flakes. This effect arises from blurring, which creates a gradation at the edge. As the contrast, which depends on the thickness, is linear from 0 to 3 layers, a gradual decrease in green channel contrast results in

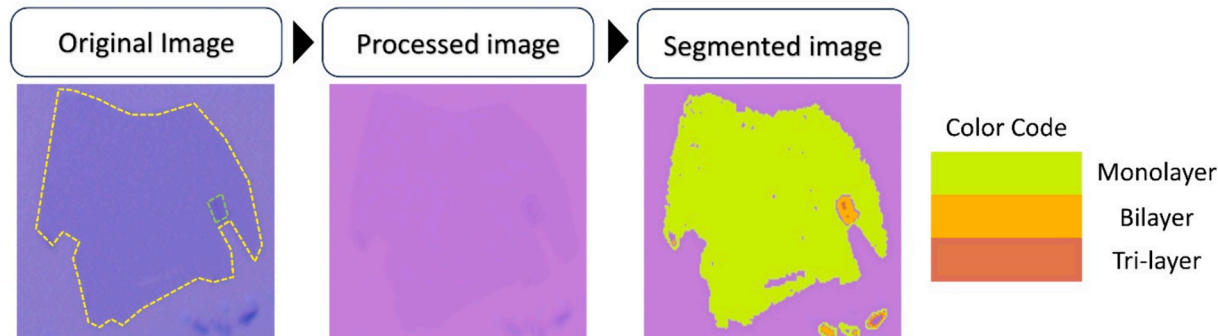


Fig. 4. An overview of the process together with color coding that represents the number of layers.

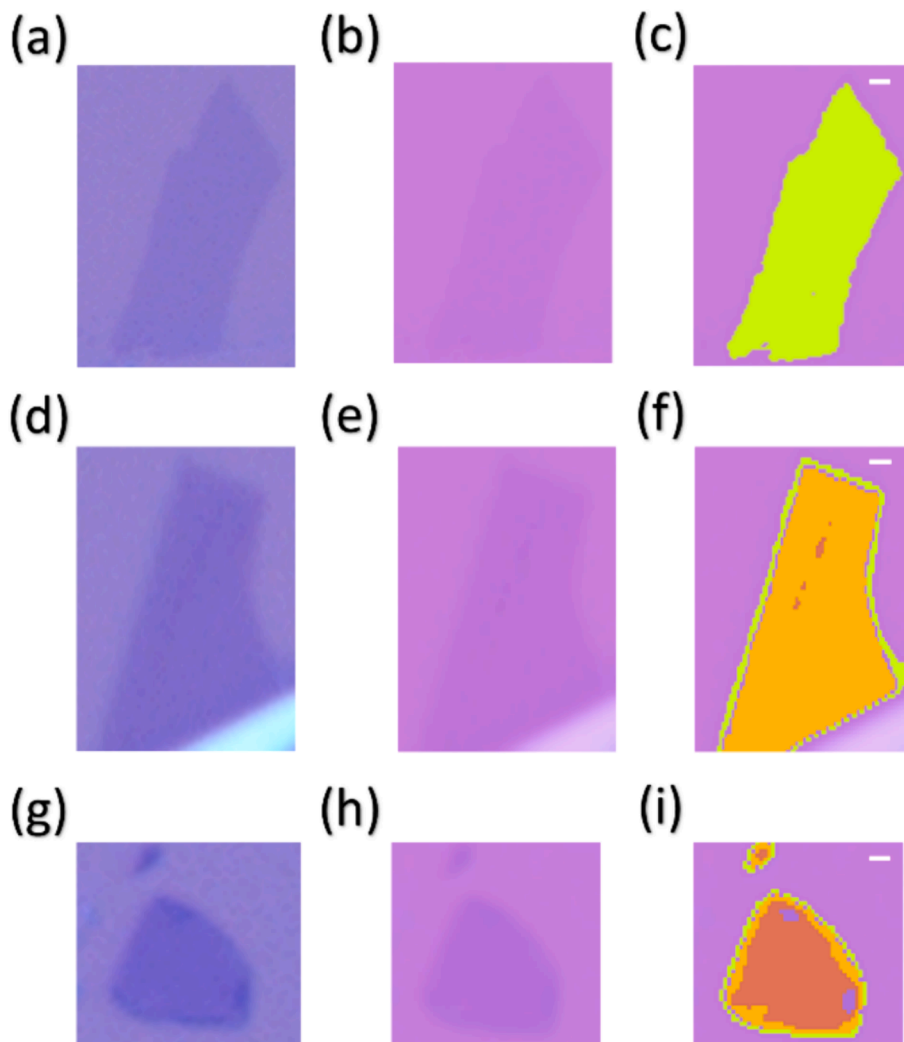


Fig. 5. Overview of the detection process. (a, d, g) Raw optical images for monolayer, bilayer and tri-layer. (b, e, h) Image after background reduction. (c, f, i) Segmented image. The scale bars correspond to 10 μm .

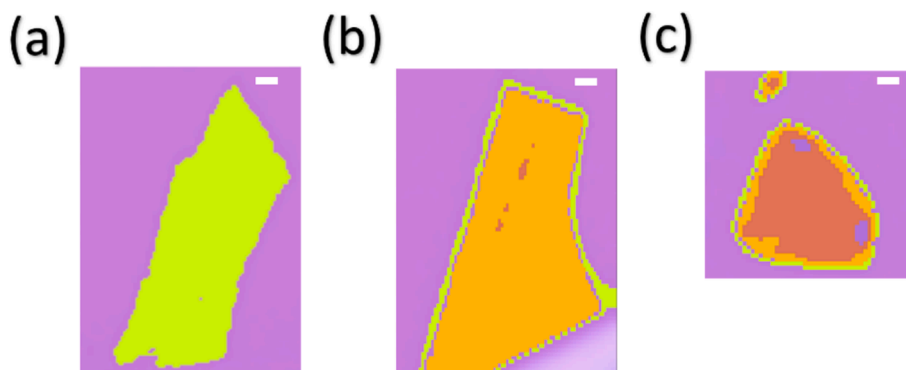


Fig. 6. (a) Monolayer shows no edge effect. (b, c) Edge effect in bilayer and tri-layer graphene. The scale bars correspond to 10 μm .

variations in layer thickness. For example, a 3-layer region may appear as 2, and a 1-layer region may appear as 0 (background) at the edge as the contrast gradually decreases. While it may initially appear counterintuitive, the edge effect serves as a noteworthy indicator of the segmentation process's effectiveness. It highlights the algorithm's ability to detect subtle changes in contrast, enabling the identification of even the most intricate image features. Fig. 6 displays images with edge effects.

3.3. Threshold tuning

To accommodate variations in environmental conditions, an intuitive GUI interface with sliders has been created for threshold adjustment ([Supplementary information](#)). Utilizing a reference image, users can fine-tune thresholds for monolayer, bilayer, and tri-layer identification. Sliding the controls adjusts the lower and upper thresholds in real-time, providing immediate feedback with a new output image. Once the

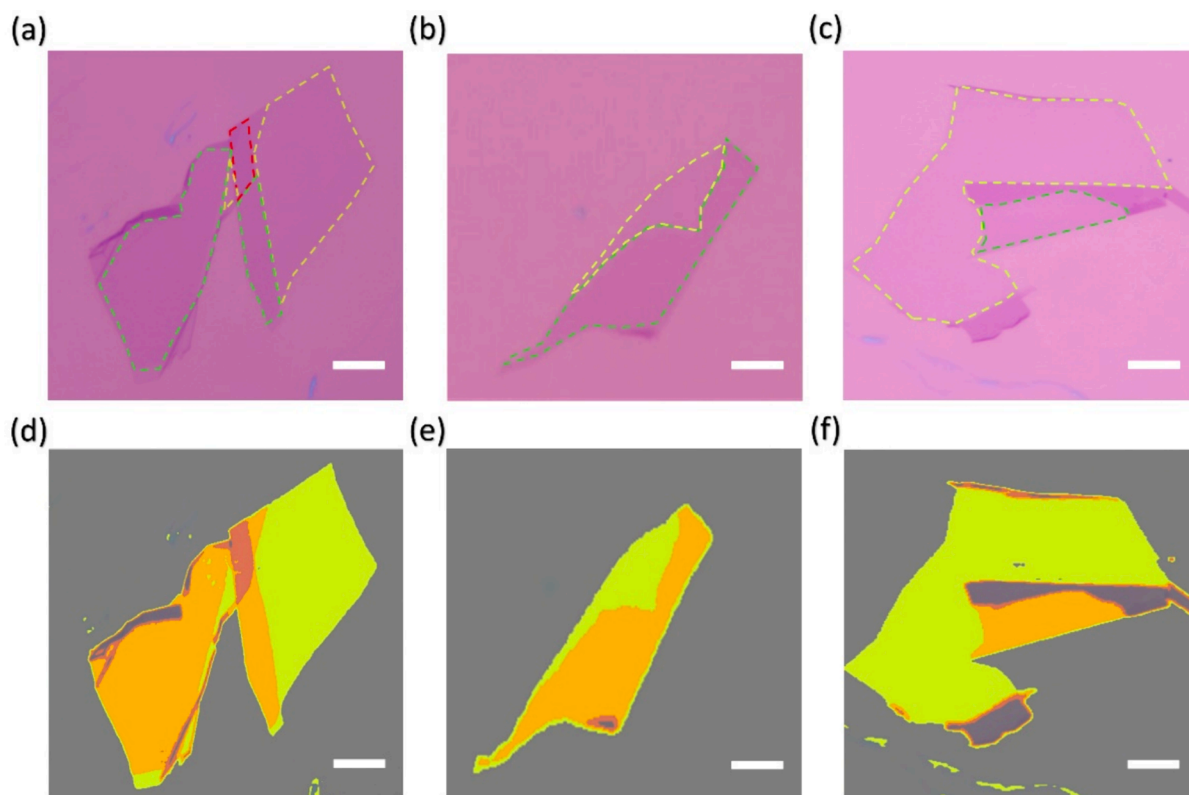


Fig. 7. (a–c) Original images of image set-2. (d–f) Segmented images after threshold tuning. The scale bars correspond to 10 μm .

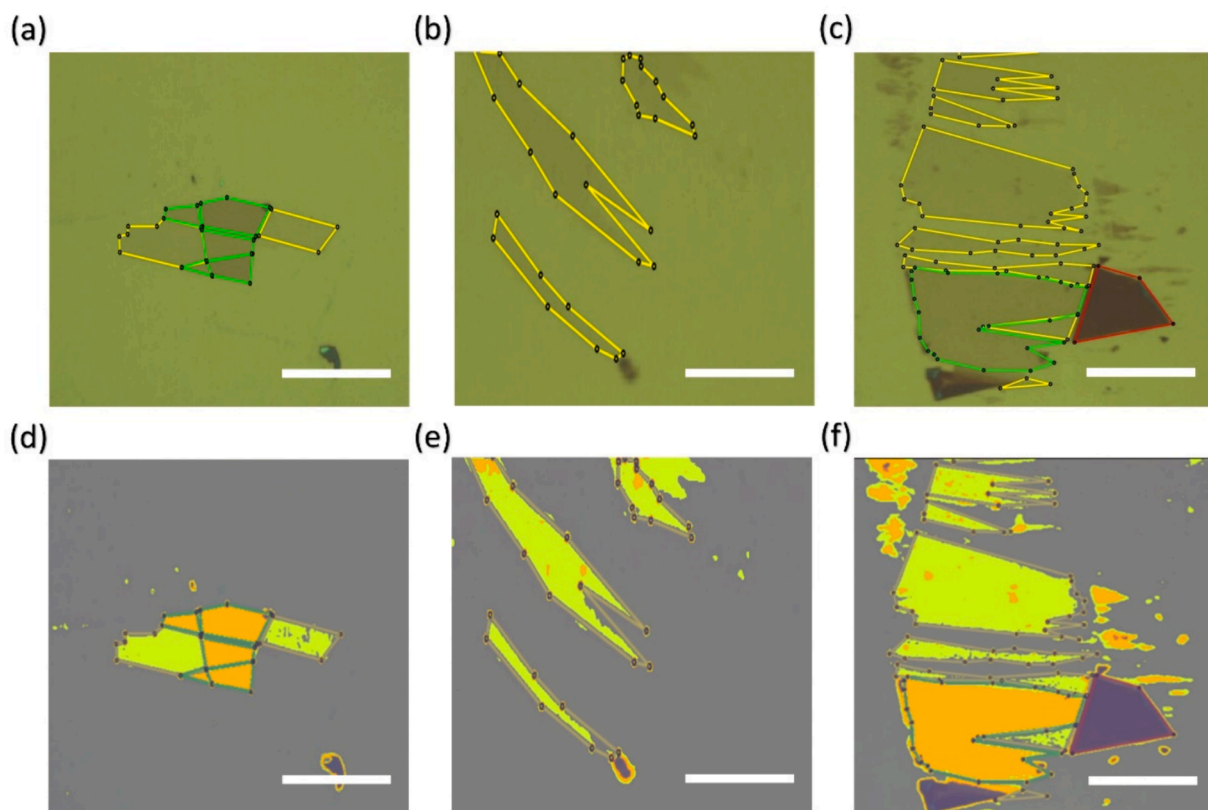


Fig. 8. (a–c) Original images of image set-3. (d–f) Segmented images after threshold tuning. The scale bars correspond to 10 μm .

Table 1
Summary of IoU accuracy for monolayer and bilayer flakes.

Layer	Image	Accuracy (%)	Average Accuracy (%)
Monolayer	1	94	94
	2	99	
	3	91	
Bilayer	1	87	89
	2	95	
	3	87	
	4	88	
	5	89	

output matches the reference, the optimized thresholds are saved for future use, ensuring accurate graphene layer identification under similar conditions.

The effectiveness of this tuning method is demonstrated by applying it to two new image sets. Set-2 images were captured using a Model CX-40M microscope from Sunny Optical Technology Company Limited at 100X magnification. Set-3 images were captured using an Asahikogaku AZ10-T/E microscope at 50X magnification. Monolayer, bilayer, and trilayer graphene are delineated by yellow, green, and red outlines, respectively. The results after adjusting the thresholds can be seen in Fig. 7 and Fig. 8.

3.4. Universality

Table 2 displays the thresholds for all three image sets used and mentioned previously. Upon comparing the thresholds developed for all three datasets, an evident pattern emerges, which underscores the efficacy of the methodology.

$T_{mono} > T_{bi} > T_{few} \text{ or } T_{many}$

This relationship can be explained as follows: when multiple layers of graphene are stacked together, the resulting optical image portrays the flake with a darker appearance. This deepening tone in the image corresponds to a reduction in the green channel value. In simpler terms, the more layers of graphene that are present, the lower the green channel value, and the thresholds corroborate this trend, serving as a reliable indicator of the number of graphene layers. It's noticeable that the monolayer threshold remains quite similar across all three datasets. Similarly, the bilayer threshold for set-1 and set-2 is also very close. This observation highlights the broad applicability of this approach. Despite variations in lighting and image conditions, the thresholds established in this paper can serve as a useful starting point. With a single reference image and the custom Python script, they can be readily adjusted. This versatility makes this methodology a more widely applicable approach for contrast-based segmentation, particularly when using a single channel.

4. Conclusion

The research presents a robust and versatile methodology for precisely identifying graphene layers in optical images by leveraging solely, the green channel value. The consistent correlation between graphene layering and corresponding green channel values further underscores the methodology's adaptability and utility across diverse datasets. These findings present a promising and universally applicable approach to contrast-based segmentation, streamlining the process while enhancing accuracy.

The observed edge effect, particularly prominent in bilayer and trilayer graphene, underscores the methodology's capacity to differentiate subtle contrasts. This event provides a dependable indicator of graphene layer count. In summary, the methodology presents a pathway to efficient and accurate graphene layer identification using a single

Table 2
Threshold values for Image Set-1, Set-2, and Set-3.

Image set	Threshold		
	Green Channel Value	% of Background	
1	Monolayer	120–122	94.5 %–96.2 %
	Bilayer	114–118	89.8 %–92.9 %
	Tri-layer	112–115	88.2 %–90.6 %
2	Monolayer	121–124	95.3 %–97.6 %
	Bilayer	115–121	90.5 %–95.3 %
	Few layers	< 115	< 95.3 %
3	Monolayer	119–122	93.7 %–96.1 %
	2–5 layers	107–118	84.3 %–92.9 %
	Many layers	< 107	< 84.3 %

channel, requiring minimal computational resources, and eliminating the need for additional instruments or setups. As a result, it offers a cost-effective solution for identifying graphene layers, making it an economically viable option for researchers and practitioners alike.

CRediT authorship contribution statement

Miah Abdullah Sahriar: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Abdul Hamid Rumman:** Writing – original draft, Visualization, Formal analysis, Data curation. **Mohd. Rakibul Hasan Abed:** Investigation, Data curation. **Ahsiur Rahman Nirjhar:** Investigation, Data curation. **Md. Nazmul Ahsan Dipon:** Investigation, Data curation. **Sadika Jan-nath Tan-Ema:** Investigation, Data curation. **Houk Jang:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Conceptualization. **Saqib Ahmed:** Writing – review & editing, Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.commatsci.2024.113198>.

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